

Tripistry: Final Robust Database Specification

COS221 Practical Assignment 5 | Future-Proofed for Core & Bonus Tasks

Design Philosophy: This specification has been strictly audited against the COS221 PA5 guidelines. It not only covers the core requirements (managing distinct roles, packages, and bookings) but proactively includes data structures to support the Bonus Tasks (Algorithmic Matching, AI Integration, Security Audits, and Interactive Maps).

1. Generalisation / Specialisation (Distinct Roles)

The spec enforces distinct roles for travellers and travel agencies, required for both the DB schema and application interface.

Entity	Type	Attributes & Bonus Task Alignment
User	Superclass	<u>UserID</u> , Email, PasswordHash, DateJoined, LastLoginTime, AccountStatus. <i>Bonus 2 (Security): LastLoginTime and AccountStatus support rate-limiting and audit logging.</i>
Traveller	Subclass	FirstName, LastName, DOB, {TravelPreferences}, SoloBudget. <i>Bonus 3 (Algorithmic Matching): Preferences and Budget allow the algorithm to match solo travellers into groups based on existing schema data.</i>
TravelAgency	Subclass	AgencyName, RegistrationNumber, {ContactNumbers}, [Address (Street, City, Zip)], <u>AverageRating</u> .

Constraint: Total and Disjoint (d). A User MUST be exactly one of the subclasses.

2. Package & Group Trip Ecosystem

Entity	Attributes & Functionality
TravelPackage	

	PackageID, Title, Description, BasePrice, TotalPrice, DurationDays, MaxCapacity, AIGeneratedSummary. <i>Bonus 4 (AI Integration): AIGeneratedSummary stores outputs from LLMs summarizing the structured package data.</i>
GroupTrip (Weak Entity)	TripDateID (Partial Key), StartDate, EndDate, Status, CurrentCapacity. <i>Parent: TravelPackage. Allows an agency to schedule the same package multiple times.</i>

3. The Travel Assets (Catalog)

These are the individual components curated into a package. To support **Bonus 5 (Interactive Map View)**, all physical locations include a Geospatial composite attribute.

Entity	Attributes
Destination	DestID , Name, Country, Region, [Coordinates (Lat, Long)] , PopularityScore.
Flight	FlightID , Airline, FlightNum, DepTime, ArrTime, [DepAirport (Code, Name)] , [ArrAirport (Code, Name)] , Cost.
Accommodation	AccommID , Name, Type, PricePerNight, StarRating, [Address] , [Coordinates (Lat, Long)] .
Attraction	AttractionID , Name, Category, EntryFee, [Coordinates (Lat, Long)] .
Restaurant	RestaurantID , Name, CuisineType, AverageCost, [Coordinates (Lat, Long)] .

4. Transactions & Feedback

Entity	Attributes & Functionality
Booking	BookingID, BookingDate, TotalAmount, PaymentStatus, PartySize. <i>Core Requirement: Links Travellers to GroupTrips/Packages.</i>
Review (Weak Entity)	ReviewID (Partial Key), Rating, Comment, DatePosted, SentimentScore. <i>Bonus 4 (AI Integration): SentimentScore allows an AI model to evaluate the text of the review.</i> <i>Parent: Traveller.</i>

5. Relationship Mapping (Chen's Notation Guidelines)

- **Creates (1:N):** TravelAgency (1) creates TravelPackage (N).
- **Schedules (Identifying 1:N):** TravelPackage (1) schedules GroupTrip (N). (Double diamond).
- **Includes_* (M:N):** Five separate M:N relationships connecting TravelPackage to Destination, Flight, Accommodation, Attraction, and Restaurant.
- **Makes_Booking (1:N):** Traveller (1) to Booking (N).
- **For_Trip (N:1):** Booking (N) is placed for a GroupTrip (1).
- **Writes_Review (Identifying 1:N):** Traveller (1) writes Review (N). (Double diamond).
- **Evaluates_Agency (N:1):** Review (N) targets TravelAgency (1).
- **Evaluates_Package (N:1):** Review (N) targets TravelPackage (1).

Instructions for the AI Coder:

Generate the XML using strict Chen's Notation. Ensure weak entities (GroupTrip, Review) have double-lined rectangles and are connected to their identifying parents via double-lined diamonds. Use dashed ovals for TotalPrice, CurrentCapacity, and AverageRating. Expand composite attributes like [Coordinates] into distinct sub-ovals for Lat and Long.